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Unit: ICS3305: Machine Learning
CAT II

Q1.

a.

i. Supervised Learning: This is a method where the model is trained on a labeled dataset, meaning the input data is paired with the correct output. The model learns to map inputs to outputs to predict labels for unseen data.

- **Example: Spam Filtering.** An email system is trained on emails labeled as "Spam" or "Not Spam". It learns features (keywords, sender info) to classify new, incoming emails automatically.

ii. Unsupervised Learning: This method involves training a model on data that has no historical labels. The system tries to learn the underlying structure, patterns, or distributions within the data without specific guidance on what the outcome should be.

- **Example: Customer Segmentation.** A supermarket analyzes purchase history to group customers with similar buying habits (e.g., "Health Conscious" vs. "Budget Shoppers") without knowing these groups in advance.

iii. Case-Based Learning (CBL): Also known as Case-Based Reasoning, this is a problem-solving paradigm that solves new problems by retrieving and adapting solutions from similar past problems stored in a database (case base).

- **Example: Help Desk Support.** When a printer malfunctions, a support system looks up past cases with similar error codes to suggest the solution that worked previously.

iv. Reinforcement Learning: This involves an agent learning to make decisions by performing actions in an environment and receiving feedback in the form of rewards or penalties. The goal is to maximize the cumulative reward over time.

- **Example: Robotic Walking.** A robot learns to walk by trying different leg movements. If it moves forward, it gets a "reward" (positive score); if it falls, it gets a "penalty," eventually learning the stable walking gait.

Q2.

a. Phases of Developing a Machine Learning Model:

1. **Problem Definition:** Understanding the business problem and determining if ML is the right solution.
2. **Data Collection:** Gathering raw data from various sources (databases, files, sensors).

3. **Data Preprocessing:** Transforming raw real-world data into clean, consistent, and suitable format. Cleaning data (handling missing values), encoding variables (converting text to numbers), removing duplicates, and normalizing/scaling features.
4. **Model Selection:** Choosing appropriate algorithms (e.g., Decision Trees, Neural Networks) based on the data type and problem.
5. **Training:** Feeding the data into the algorithm to create the model.
6. **Evaluation and fine-tuning:** Testing the model using metrics (Accuracy, Precision) on unseen data, adjusting hyper-parameters and retraining to get to desired metrics levels.
7. **Deployment:** Integrating the model into a production environment to make real-world predictions.
8. **Monitoring & Maintenance:** Tracking performance over time and retraining if the data drifts.

b.

i. Algorithm Choice & Justification: I would choose **Decision Trees** and **K-Nearest Neighbors (KNN)**.

- **Decision Tree:** It is highly interpretable. For a social problem like divorce, understanding *why* a prediction was made (e.g., "Income < 5 implies Divorce") is valuable. It also handles the mix of numerical (Years, Income) and categorical data well.
- **K-Nearest Neighbors (KNN):** It is a simple, instance-based learning algorithm that works well with small datasets like the one provided (10 rows). It predicts based on the similarity to other families, which is intuitive.

ii. See attached separate file for implementation

iii. **Recommendation: K-Nearest Neighbors (KNN)**

Reasoning:

1. **Robustness to Data Scarcity (The "Small Data" Caveat):** The primary reason for choosing KNN is its ability to handle extremely small datasets better than Decision Trees. With only 8 samples in the training set, the Decision Tree likely suffered from **overfitting**. It memorized the specific noise or exact patterns of those 8 families, creating rigid rules that failed completely (0% accuracy) when faced with new, unseen data. KNN, being an "instance-based" learner, does not try to learn a global rule; it simply looks for the most similar existing examples, which is a safer approach when data is too scarce to generalize broad patterns.
2. **Empirical Performance:** In this experiment, KNN achieved **50% accuracy**, whereas the Decision Tree achieved **0%**. In any practical application, a model that provides some correct predictions is infinitely better than one that gets everything wrong, regardless of other features like interpretability.
3. **Stability:** Decision Trees are known to be "high variance" algorithms, meaning a tiny change in the training data (like having 8 rows instead of 9) can result in a completely different tree

structure. KNN is generally more stable in these low-data environments because it relies on local averaging rather than hierarchical splitting.

Note on Trade-offs: While Decision Trees are usually preferred for divorce prediction due to **explainability** (understanding *why* a marriage fails), explainability is essentially useless if the model's predictions are incorrect. Therefore, given the constraints of this specific dataset, KNN is the only viable choice.

Q3.

a. Terms Description:

- **i. Accuracy:** The ratio of correctly predicted observations (both True Positives and True Negatives) to the total observations. It measures how often the model is correct overall.
- **ii. Precision:** The ratio of correctly predicted positive observations to the total predicted positives. It answers: "Of all the instances defined as Positive, how many were actually Positive?"
- **iii. Sensitivity (Recall):** The ratio of correctly predicted positive observations to the all observations in the actual class. It answers: "Of all the actual Positive instances, how many did we predict correctly?"
- **iv. ROC Curve:** A graphical plot that illustrates the diagnostic ability of a binary classifier system. It plots the **True Positive Rate (Sensitivity)** against the **False Positive Rate (1 - Specificity)** at various threshold settings.

b. Chick Gender Prediction Analysis

Part a: Compare models based on Accuracy

We calculate accuracy using the formula: $\text{Accuracy} = (\text{TP} + \text{TN}) / \text{Total Samples}$

- Model 1 (Naive Bayes):
 - True Positives (Male predicted as Male): 35
 - True Negatives (Female predicted as Female): 30
 - False Positives (Female predicted as Male): 10
 - False Negatives (Male predicted as Female): 25
 - Total Samples: $30 + 10 + 25 + 35 = 100$
 - **Accuracy:** $(30 + 35) / 100 = 65 / 100$
 $= 65\%$
- Model 2 (Decision Tree):
 - True Positives (Male predicted as Male): 32
 - True Negatives (Female predicted as Female): 28

- False Positives (Female predicted as Male): 12
- False Negatives (Male predicted as Female): 28
- Total Samples: $28 + 12 + 28 + 32 = 100$
- **Accuracy:** $(28 + 32)/100 = 60/100$
= 60%

Conclusion: Model 1 (Naive Bayes) is the better model with an accuracy of 65% compared to Model 2's 60%.

Part b: Measures for the Best Fitting Model (Model 1) *Assumption: "Male" is treated as the Positive Class.*

- **i. Sensitivity (Recall):**

Formula: $TP/(TP + FN)$

Calculation: $35/(35 + 25) = 35/60$
= 58.3%

- **ii. Specificity:**

Formula: $TP/(TP + FP)$

Calculation: $30/(30 + 10) = 30/40$
= 75%

- **iii. False Positive Rate (FPR):**

Formula: $FP/(TN + FP)$ or $(1 - \text{specificity})$

Calculation: $10/(30 + 10) = 10/40$
= 25%

Q4.

Four issues that arise in high-stakes fields (Healthcare, Finance, Law, Agriculture) include:

1. **Bias and Fairness:** If the training data contains historical biases (e.g., medical or finance data heavily skewed toward one demographic), the model will replicate these biases, potentially leading to discriminatory loan denials or incorrect medical diagnoses for minority groups.
2. **Explainability (The "Black Box" Problem):** In law and medicine, it is not enough to have a prediction; professionals need to know *why*. Complex models like Deep Learning often cannot explain their reasoning, making them difficult to trust in life-or-death decisions.
3. **Data Privacy and Security:** High-stakes applications often use sensitive personal data (medical records, financial history). There is a risk of data breaches or models inadvertently revealing private information (inference attacks).
4. **Robustness and Reliability:** In agriculture or autonomous finance, models must handle unexpected scenarios (e.g., extreme weather, market crashes). A model that works well in a lab but fails under "noisy" real-world conditions can cause catastrophic financial loss or crop failure.